AAYUSH PANDEY

+91 9315905384 | aayush4373pandey@gmail.com | LinkedIn | GitHub

Education

VIT Bhopal University **August 2027**
Bachelor of Technology in Computer Science and Engineering *Madhya Pradesh*

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL
Frameworks & Runtime: React, Express.js, RESTful APIs, Tailwind CSS, Electron.js, Node.js
Databases: MySQL, Oracle SQL, MongoDB, SQLite
Tools & Platforms: Git, GitHub, VS Code, Vercel, AWS, Docker, Vite
ML & Data: TensorFlow, Keras, OpenCV, pydicom, scikit-image, NumPy
Core Competencies: System Design, UI/UX Design, OOP Principles, Data Structures & Algorithms, Operating Systems, Linux

Projects

MindConnect |Github| TypeScript, React, Node.js, Prisma, PostgreSQL (Neon), Vercel

- Architected a full-stack mental health platform with 10+ REST API endpoints across admin, auth, booking, and doctor management modules
- Designed a PostgreSQL schema reducing query redundancy and improving data consistency across 5 core modules using Prisma ORM for type-safe database access
- Implemented JWT-based authentication, securing 100% of user routes and enabling role-based access control for admin operations.
- Deployed on Vercel with Neon PostgreSQL, using a monorepo architecture with shared TypeScript types across client and server.

MediScan |Github| Python, Electron.js, OpenCV, SQLite, pydicom, scikit-image

- Built a cross-platform medical imaging desktop application to process DICOM and image formats for automated fat and muscle tissue segmentation using a 7-stage Python ML pipeline
- Applied advanced image processing techniques to improve tissue detection accuracy and reliability (30px edge filter, 50px minimum region threshold)
- Developed an interactive DICOM editing tool enabling region-of-interest selection and export with preserved metadata and auto-generated SOP Instance UIDs
- Engineered a SQLite-based storage system to manage 10+ clinical metrics
- Generated 5 visualization outputs per analysis with in-app canvas-based editing, undo/redo, and result export, enabling faster clinical interpretation and improving result interpretability for end users.

Plant Disease Detection | Python, OpenCV, TensorFlow, HTML, CSS, JavaScript

- Built an end-to-end plant disease detection system by training a custom CNN on 87,000+ images across 38 classes, achieving 99%+ validation accuracy
- Designed a browser-based UI using HTML, CSS, and JavaScript to upload leaf images and display real-time predictions
- Integrated the ML model with the frontend via a Python-based inference pipeline, handling image preprocessing (OpenCV) and prediction flow
- Applied data augmentation and optimized the model for lightweight inference, enabling faster response times in the user interface.